Hyangsuk Min

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Research Profile

My research asks how AI agents can remain trustworthy when the information they receive is incomplete, noisy, evolving, or distributed across multiple contexts. I develop data-centric evaluations that examine how models perceive, structure, remember, and use information, diagnosing failures in evidence use, hallucination, and consistency. This perspective extends naturally to memory-enabled agents, where prior interactions and task experience must be retained and reused without compromising faithfulness. I also apply this lens to AI co-scientist systems, evaluating whether LLMs exhibit researcher-level understanding in scientific literature reading, comparison, synthesis, and hypothesis generation.

Education

- **Korea Advanced Institute of Science and Technology** — *Sep 2024 - Present*
  Department of Industrial System Engineering (advisor : Prof. Hwanjun Song) — Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology** — *Mar 2020 - Aug 2022*
  M.S., Graduate School of Data Science (advisor : Prof. Jae-Gil Lee) — Daejeon, South Korea
- **Kookmin University** — *Mar 2015 - Aug 2019*
  B.S., Bigdata Anaysis and Statistics — Seoul, South Korea

Publications

NLP Min, H*., Song, H. (2026). Don't Scroll Back: Missing-Evidence Memory for Streaming Dialogue Summarization (under review)

NLP Yun, T*., **Min, H*.**, Lee, Y., Bang, S., Song, H. (2026). Designing Human-Multi-Agent Collaboration for Fact-checking in Crowdsourced Annotation Workflows (under review)

Ban, M*., Choi, J*., **Min, H*.**, Kim, NH., Song, H. (2026). Completing Missing Annotation: Multi-Agent Debate for Accurate and Scalable Relevance Assessment for IR Benchmarks. In Proc. *The Fourteenth International Conference on Learning Representations (ICLR)*

Min, H., Song, H. (2025). Streaming Dialogue Summarization: A Memory-Augmented Framework for Summarization and Evaluation. In Korea Software Congress (KSC).

Ban, M*., Choi, J*., **Min, H*.**, Kim, NH., Bang, S., Song, H. (2025). BRIDGE: Toward Faithful RAG Benchmarking via Retrieval-Generation Alignment. In Proc. *the 34th ACM International Conference on Information and Knowledge Management Workshop (CIKM Workshop)*.

Lee, Y*., Deng, J*., Kim, NH., **Min, H.**, Yun, T., Ban, M., Kim, Y., Song, H. (2025). Towards a Holistic and Automated Evaluation Framework for Multi-Level Comprehension of LLMs in Book-Length Contexts. In Proc, *the 2025 Conf. on Empirical Methods in Natural Language Processing (EMNLP)*.

Yun, T., Oh, J., **Min, H.**, Lee, Y., Bang, J., Cai, J., Song, H. (2025). ReFeed: Multi-dimensional Summarization Refinement with Reflective Reasoning on Feedback. In Proc, *Conf on Language Modeling (CoLM)*.

Min, H*., Lee, Y*., Ban, M., Deng, J., Kim, NH., Yun, T., Su, H., Cai, J., Song, H. (2025). Towards Multi-dimensional Evaluation of LLM Summarization across Domains and Languages . In Proc. *the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Min, H, Lee, J. (2023). Temporal Convolutional Network-Based Time-Series Segmentation. In Proc, *2023 IEEE Int'l Conf. on Big Data and Smart Computing*.

Kim, D., **Min, H.**, Nam, Y., Song, H., Yoon, S., Kim, M, and Lee, J. (2022). COVID-EENet: Predicting Fine-Grained Impact of COVID-19 on Local Economies. In Proc. *36th AAAI Conf. on Artificial Intelligence (AAAI)*.

Kim, M., Kang, J., Kim, D., Song, H., **Min, H.**, Nam, Y., Park, D. and Lee, J., (2020). Hi-COVIDNet: Deep Learning Approach to Predict Inbound COVID-19 Patients and Case Study in South Korea. In *Proc. 26th ACM SIGKDD Int'l Conf. on Knowledge Discovery and Data Mining (KDD)*.

Professional Experience

- **M2MCx, SW Campus, HL Mando.** *Oct 2022 - Aug 2024*
  AI Research Engineer Pangyo, South Korea
  - Built auto-ML training, evaluation, and deployment pipelines for smart factory and vehicle diagnostics systems.
  - Developed on-premise MLOps infrastructure and CI/CD workflows for deep learning models using sensor and image data in production environments.
  - Designed synthetic data generation pipelines for vision inspection and uncharted driving scenarios under limited-data and distribution-shift settings.
  - Applied prompt engineering and domain adaptation to LLM-based OEM requirement classification and mixed-format document analysis.
  - Worked across the full ML lifecycle, including data analysis, model development, evaluation, deployment, and production monitoring.
- **Physical Design, Uniquify** *Aug 2018 - June 2019*
  ASIC Engineer Intern San Jose, CA, USA
  - Worked with overseas engineering teams on physical design, PnR, and engineering change order workflows.
  - Developed Perl scripts to summarize timing reports and identify design violations.

Honors and Awards

- **Graduated Cum Laude** *Aug 2019*
  Kookmin University
  - GPA: 4.4/4.5
- **Merit scholarship** *Sep 2015-Mar 2019*
  Kookmin University
  - Awarded merit scholarships totaling approximately $11,000 across 7 semesters for academic excellence
  - 2 full, 5 partial scholarships.

Certifications

- **ADP: Advanced Data Analytics Professional** *07 2021*
- **SQLD: SQLDeveloper** *04 2018*

Patent

- **Method and Apparatus for Predicting Fine-Grained Impact of COVID-19 on Local Economies** *Oct 19, 2022*
  Registration Number: 1020220134903 South Korea

Skills

- **Programming Languages:** Python (Pytorch), R
- **Web Technologies:** CSS, HTML
- **Database Technologies:** SQL
- **Languages:** Korean (Native), English (Intermediate)

Activities

- **Reviewer:** ICML, ACL, EMNLP, NeurIPS, ICLR, Neural Networks

Teaching Experience

- **LLM Fine-tuning with HuggingFace (Assistant instructor, SKT):** Nov, 2023
- **Data Structures and Analysis (IE260 lecture TA, KAIST):** Spring, 2025
- **Programming for Data Science (IE260 lecture TA, KAIST):** Spring, 2026

Updated as of January 27, 2026